Day 1. MODULE: PYTHON ESSENTIALS INTRODUCTION

What is Python...?

Python is a high-level, interpreted programming language known for its

1) simplicity,
2) versatility,
3) readability.
4) Community and support.

It features general purpose which is clean and concise syntax, making it easy to learn and use for a wide range of tasks, from

1) Web development
2) Data analysis
3) Artificial intelligence (ML, DL)
4) Scientific Computation
5) Game Development

Python emphasizes code readability and encourages a modular and object-oriented programming approach, fostering rapid development and collaboration.

A Brief History of Python

Python was created by **Guido van Rossum** in the late **1980s,** currently there are three version of python

- 1st release in **1991.**
- **2nd Python 2.0 in 2000**
- **3rd release in 2008 Python 3.0 in,** It underwent major revisions with aiming for simplicity, readability, and efficiency, evolving into one of the most popular programming languages globally with a rich ecosystem and supportive community.



Installing Python
https://www.python.org/downloads/

Installing Anaconda
https://www.anaconda.com/download

Why anaconda Navigator?(jupyter)

Anaconda is preferred for Python development because it offers

1) **easy package management with Conda**, includes a wide range of
2) **pre-installed libraries** and tools, is compatible across multiple platforms, provides performance optimization, and simplifies the deployment of Python applications.

Python spefic version
Pandas
Numpy
Matplotlib

Vs code

Xyz app
Env(package)3.7 pythond, pandas1.2 , numpy1.2

Abc app
Env(package)3.8 pythond, pandas1.2 , numpy1.2,

